

Spending Money Earned by Someone Else on Yourself

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Abstract

People frequently spend money earned by others on themselves. Children spend their parents' money on groceries; romantic partners draw from joint accounts; employees charge business meals to expense accounts. Yet how the source of money affects spending behavior and its psychological experience remains largely untested. This paper investigates how self-earned versus partner-earned money affects willingness to pay and the pain of paying when individuals make personal purchases, using social distance between spender and earner as the organizing theoretical variable.

We develop a theoretical framework in which two competing forces — a windfall benefit from spending unearned funds and a guilt cost from potentially disadvantaging the earner — shape optimal consumption. Social distance governs their relative strength, generating an ambiguous prediction at low social distance and an unambiguous positive effect of other-earned money on consumption at high social distance. To test these predictions, we design two complementary laboratory experiments: a couples experiment using romantic partners to anchor the low end of the social distance spectrum, and a stranger extension using randomly matched strangers to anchor the high end.

Preliminary results from 100 couples (71% of the target sample) find no significant average difference in willingness to pay across conditions, consistent with offsetting windfall and guilt effects at low social distance. Pain of paying, however, is substantially higher when spending partner-earned money than self-earned money. The divergence between these outcomes — unchanged behavior alongside a sharply different psychological experience — motivates the stranger extension as a direct test of whether behavioral effects emerge at greater social distance.

Keywords: Pain of paying; spending; social preferences; social distance; source of money; someone else's money

JEL Classification: C91; D11; D14; D16; D31; D91

1. Introduction

There are many circumstances in which people spend other people's money on themselves. Children purchase groceries using their parents' money. Spouses spend money from shared joint accounts. Employees take business partners to dinner on departmental funds. Policymakers spend taxpayer dollars on legislation. But how differently do they act than if they were spending their own?

Friedman and Friedman (1990) suggest that people have no strong incentive to economize when spending someone else's money on themselves and will be more wasteful. This paper empirically investigates how the source of money — self-earned versus earned by someone else — affects people's spending on themselves. Spending another person's money can carry its own psychological costs. People have strong preferences for fairness and reciprocity (Fehr and Schmidt, 1999), and spending money earned by someone else may trigger guilt and heightened feelings of responsibility. These forces push in opposite directions. Spending unearned funds generates a windfall benefit, while spending money earned by someone psychologically close generates guilt and a sense of responsibility. How strongly each force operates likely depends on the social distance between the spender and the earner: how psychologically close they are to the person whose money they are spending.

Further, social distance might matter to the net outcome of these effects. At low social distance, as between romantic partners, windfall benefits and guilt costs may roughly offset each other, leaving the net effect on spending an empirical question. At high social distance, as between strangers, guilt diminishes and windfall benefits grow, and the net effect on spending is predicted to be unambiguously positive.

We develop a theoretical framework that formalizes how the windfall benefit and guilt cost of spending another person's money affect spending, and how both depend on social distance to the earner. To empirically examine how spending varies across self-earned and other-earned money, we design two complementary laboratory experiments: a couples experiment, currently underway, and a stranger extension, in which participants are paired with randomly matched strangers, anchoring the design at both ends of the social distance spectrum. Our primary outcome is participants' willingness to pay for real consumer products, elicited using an incentivized auction; our secondary outcome is anticipated

pain of paying, rated across a range of hypothetical price points using a validated scale.¹

Preliminary results from the couples experiment, based on 100 of a target 140 couples, speak to both outcomes. In the experiment, one member of each couple bid in an incentivized auction for a consumer product, once using money they earned and once using money their partner earned. Money source does not significantly affect willingness to pay: spenders bid similarly whether spending self-earned or partner-earned money, a null result the model does not rule out, as no directional prediction follows at low social distance. Among the 39 spenders who bid differently across conditions, 24 (62%) bid more with self-earned money, and the mean difference within this subset is marginally significant (Wilcoxon $p = 0.071$); the null in the full sample is driven largely by the 61% of spenders who bid identically across conditions.

Pain of paying tells a different story: spenders report consistently higher pain when spending their partner's money than their own, a medium-sized effect (Cohen's $d = 0.518$) that holds across parametric and non-parametric tests. This divergence between behavior and psychological experience is itself a finding. The stranger extension tests whether the behavioral null disappears when guilt diminishes and windfall benefits grow at greater social distance.

Section 2 reviews the relevant literature on mental accounting, windfall effects, fairness preferences, the pain of paying, and social distance. Section 3 develops a theoretical framework to formalize the windfall-guilt tradeoff and derive its comparative static across social distance. Section 4 details the experimental design for both the couples experiment and the planned stranger extension. Section 5 presents preliminary results. Section 6 concludes with interpretations, limitations, and directions for future work.

2. Background

No prior research has directly tested whether people spend other people's money more wastefully than their own; the relevant literatures span mental accounting, windfall effects, fairness preferences, and social distance. When making spending decisions, people use "mental accounting" to categorize and evaluate choices by grouping money into separate mental accounts (Thaler, 1985; Prelec and Loewenstein, 1998). Thaler (1999) argues that

1. Prior survey evidence from Enriquez et al. (2019) finds that pain of paying is higher when spending a domestic partner's earnings than one's own; we build on that finding with an incentivized laboratory design and extend it across the social distance spectrum.

because of these accounts, people do not treat money as fungible, instead assigning explicit and implicit labels based on source (regular income versus windfalls) or intended purpose. People may therefore assign self-earned and other-earned money to different mental accounts, spending each differently even when the nominal value is identical.

People also tend to be more conservative and self-interested when spending money they earned than money they received as a windfall (Cherry et al., 2002). If people treat money earned by another person as a windfall, they may spend it more wastefully than money they earned themselves. The strength of this windfall effect likely depends on the psychological distance between spender and earner: at greater distance, the earner's effort becomes less salient, and the money feels more like an unearned transfer than the product of someone else's work.

While these factors suggest more liberal spending with other-earned money, additional factors may push in the opposite direction. People value equitable outcomes beyond narrow self-interest (Fehr and Schmidt, 1999). When people spend other-earned money on themselves, they may self-regulate out of internalized notions of fairness and reciprocity. Even when the person holds legal property rights to the money, not having earned it themselves may generate feelings of unfairness (Oxoby and Spraggon, 2008; Dannenberg et al., 2012). This fairness perspective suggests that spending is more restrained with other-earned than with self-earned money. How strongly these fairness concerns operate likely depends on how close the spender is to the earner, a pattern that may similarly govern the pain of paying.

Another potential cost from spending money is the pain of paying. fMRI evidence shows that anticipating and experiencing payment activates brain regions associated with physical pain processing (Mazar et al., 2016). The pain of paying is a proxy for the opportunity cost of consumption, reducing willingness to pay and helping regulate spending decisions (Prelec and Loewenstein, 1998; Loewenstein and O'Donoghue, 2006; Rick et al., 2008). The source of money may directly affect the pain of paying. The Friedman and Friedman (1990) conjecture that people spend other people's money more wastefully implicitly assumes that pain of paying is lower when spending money earned by someone else, since otherwise overspending would not follow. People may experience lower pain of paying when spending money they did not earn, spending more wastefully as a result. Alternatively, the pain of paying may be higher with other-earned money, because of increased feelings of responsibility or guilt. Whether pain of paying rises or falls with other-earned money

may therefore depend on the same factor: how psychologically close the spender is to the earner.

One key determinant of the pain of paying could be social distance. Social distance — the sense of psychological connection or isolation between people — affects economic decision-making: in a dictator game, greater social distance between dictator and recipient led to more self-interested behavior (Hoffman et al., 1994). Fiedler et al. (2011) extend this finding, showing that social distance affects both generosity and risk-taking. Social distance may similarly affect the pain of paying when spending other-earned money. When the money comes from someone psychologically close, such as a family member or close friend, people may feel stronger pain of paying because of increased feelings of responsibility, potential regret, or anticipated social consequences. Conversely, when the source is socially distant, such as a stranger or a large institution, pain of paying may be diminished, producing more liberal spending.

Enriquez et al. (2019) conduct two surveys to test whether pain of paying is higher when spending self-earned versus partner-earned money. The first survey uses a between-subjects design in which participants rate their feelings about a \$300 smartphone purchase with self-earned versus partner-earned money. Pain of paying is measured using the smiley Likert scale shown in Figure 1, widely used in healthcare settings to assess pain intensity.

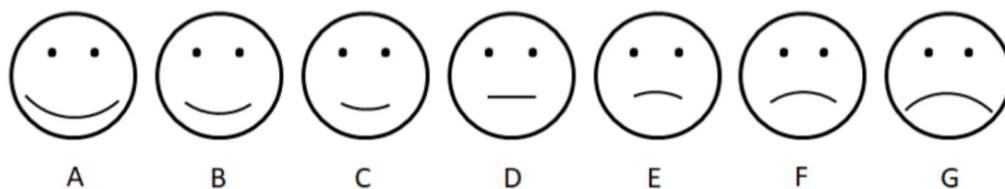


Figure 1: Smiley Likert scale used to assess pain of paying

The second survey uses a between-subjects design in which participants rated their feelings about a hypothetical \$300 purchase of either a necessity (work clothes) or an indulgence (brand-name leisurewear). Across both surveys, pain of paying was higher when spending a partner's earnings than one's own. The Enriquez et al. (2019) surveys established pain of paying as a potential mechanism to moderate spending behavior. Windfall effects, fairness concerns, and the pain of paying can each push in opposite directions, and their net effect depends on money source and the spender's social distance to the earner.

Section 3 develops a theoretical framework that formalizes these forces and derives their implications for optimal consumption.

3. Theoretical Framework

The source of money affects optimal consumption through two competing psychological forces. If people treat other-earned money as a windfall, they may spend more freely. If fairness preferences are strong and they feel guilt from spending other-earned money, they may spend less freely. Pain of paying with another person's money may be higher or lower, due to the counteracting psychological windfall benefits and guilt costs pushing in opposite directions, and the net effect likely depends on how close the spender is to the earner. We develop a framework to model consumer behavior when spending decisions involve money earned by another person, building on Thunström and Jones Ritten (2019).

Two competing forces shape the consumer's decision when spending other-earned money. First, spending money one did not earn reduces the felt opportunity cost of a purchase, generating a windfall benefit from spending unearned funds. Second, when the money was earned by someone psychologically close, spending it introduces guilt and fairness concerns, generating a guilt cost from potentially disadvantaging the earner. The relative strength of these forces determines whether a consumer spends more or less freely with other-earned money than with self-earned money.

The core setup compares two problems. In the self-earned benchmark, the consumer faces a standard opportunity cost of spending money from a joint account. In the other-earned case, two competing psychological forces are added: a windfall benefit from spending unearned money and a guilt cost from potentially disadvantaging the earner. Social distance between the consumer and the earner governs the relative strength of these forces, generating predictions about both the level of other-earned spending relative to self-earned spending and how that gap changes with social distance. The model treats the consumer's problem in isolation, abstracting away from strategic interaction between spender and earner, as the experimental design does not allow the earner to observe or respond to the spending decision in real time. At low social distance, the framework yields no general directional prediction for spending behavior or pain of paying. At high social distance, it predicts unambiguously higher consumption with other-earned money, motivating both the couples experiment and the planned stranger extension.

3.1. Framework

Consider a representative consumer who derives utility from consumption in two scenarios: spending self-earned money and money earned by another person (other-earned money). In each scenario, earnings are deposited into a shared “joint” account over which both partners hold equal property rights; the spender may spend any amount up to the full account balance, and any unspent funds are split equally between the spender and the earner. The model incorporates a joint account for two reasons. First, it equalizes property rights. When the earner retains full rights to unspent money, the spender may perceive spending other-earned money as appropriating someone else’s resources rather than making a genuine consumption decision, conflating concerns about legal ownership with the psychological cost of spending.

Second, it holds opportunity cost constant across scenarios: a dollar spent is a dollar not split, regardless of who earned it. Without this assumption, the opportunity cost of spending other-earned money would generally be lower than that of self-earned money: unspent funds typically revert to the earner rather than the spender, and spenders may feel less entitled to the remainder in any case. That difference in opportunity cost would push toward higher spending with other-earned money through a channel separate from windfall and guilt, obscuring the psychological effect of interest. The joint account assumption eliminates this channel, ensuring that any difference in optimal consumption across scenarios reflects the psychological effect of money source alone, rather than differences in property rights or opportunity cost. Section 4 describes how the experimental design implements this assumption.

The consumer derives positive utility $u(x)$ with diminishing returns from consuming quantity x , where utility depends only on the quantity consumed and not on the source of funds ($u' > 0, u'' < 0$). When spending self-earned money, the consumer faces the pain of paying $p_s(x)$, which acts as a proxy for the opportunity cost of consumption. We model the opportunity cost of self-earned spending as $c(x)$, which also serves as the pain of paying in this scenario, so $p_s(x) = c(x)$. The opportunity cost of spending is positive and strictly convex ($c' > 0, c'' > 0$). The consumer then solves

$$\max_x U = u(x) - c(x) . \tag{1}$$

The first-order condition for optimal self-earned consumption x^* is

$$\underbrace{u'(x^*)}_{\text{Marginal benefit}} = \underbrace{c'(x^*)}_{\text{Marginal opportunity cost}} \quad (2)$$

At the optimal level of consumption, the marginal benefit of consumption equals the marginal opportunity cost — which, in the self-earned case, is also the marginal pain of paying.

The other-earned problem departs from this benchmark by adding two competing psychological forces: a windfall benefit and a guilt cost, both of which depend on consumption x and social distance s . Social distance $s \in [0, 1)$ captures the degree of psychological separation between the consumer and the earner, with $s = 0$ representing oneself; values closer to 1 represent greater psychological separation. A romantic partner corresponds to low s ; a randomly matched stranger corresponds to high s . At low social distance, guilt costs are relatively strong and windfall benefits relatively weak; at high social distance, this relationship reverses.

Consistent with evidence that people exhibit greater concern for those who are psychologically close (Hoffman et al., 1994; Fiedler et al., 2011), the windfall benefit $w(x, s)$ captures the psychological benefit of spending unearned funds, increasing in both consumption and social distance ($w_x > 0, w_s > 0$). A consumer who is more socially distant from the earner focuses more on the benefit of spending money that is not their own and less on potential consequences for the earner. The guilt function $g(x, s)$ captures the psychological cost of spending another person's resources, increasing in consumption but decreasing in social distance ($g_x > 0, g_s < 0$). Guilt is stronger when the earner is psychologically close, because the spender feels greater responsibility toward them.

In the other-earned money scenario, the consumer's pain of paying is equal to the opportunity cost of consumption plus the net effect of the windfall benefit and the guilt cost, $p_o(x) = c(x) + g(x, s) - w(x, s)$. The consumer's problem becomes

$$\max_x U = u(x) - c(x) + w(x, s) - g(x, s), \quad (3)$$

with first-order condition

$$\underbrace{u'(x^*)}_{\text{Marginal benefit}} + \underbrace{w_x(x^*, s)}_{\text{Marginal windfall benefit}} = \underbrace{c'(x^*)}_{\text{Marginal opportunity cost}} + \underbrace{g_x(x^*, s)}_{\text{Marginal guilt cost}}. \quad (4)$$

At the other-earned optimum, marginal benefit plus marginal windfall benefit equals marginal opportunity cost plus marginal guilt cost. Second-order conditions for an interior solution require that windfall benefit be concave in consumption ($w_{xx} < 0$) and guilt cost be convex ($g_{xx} > 0$); these assumptions are discussed further in Appendix A.

3.2. Predictions

Let x_s^* and x_o^* denote optimal consumption under the self- and other-earned problems. The first-order conditions for optimal consumption under each problem are

$$\begin{aligned} u'(x_s^*) &= c'(x_s^*) \\ u'(x_o^*) + w_x(x_o^*, s) &= c'(x_o^*) + g_x(x_o^*, s). \end{aligned}$$

Subtracting the self-earned condition from the other-earned condition reveals what drives divergence between the two optima:

$$[u'(x_o^*) - c'(x_o^*)] = g_x(x_o^*, s) - w_x(x_o^*, s). \quad (5)$$

The left side is the gap between marginal benefit and marginal opportunity cost at x_o^* . At the self-earned optimum this gap is zero by Equation (2), so any departure of x_o^* from x_s^* is accounted for entirely by the right side. The right side measures how much marginal guilt exceeds marginal windfall. Because the joint account holds $c(x)$ constant across money sources, any observed differences in behavior between money sources reflect $g_x - w_x$ alone. The sign of $g_x - w_x$ is ambiguous because it depends on the relative magnitudes of the two forces, which is an empirical question. At low social distance both forces are active, and no directional prediction follows.

At greater social distance, the balance between these forces shifts unambiguously toward the windfall benefit. The marginal guilt cost of consumption falls as the earner becomes more distant ($g_{xs} < 0$) because a consumer who feels less responsibility toward the earner bears less guilt per unit spent. The marginal windfall benefit rises correspondingly ($w_{xs} > 0$) because greater distance reduces attention to consequences for the earner and

increases the felt gain from spending unearned funds. These cross-partial assumptions follow directly from the first-order properties of g and w : a weaker guilt function at greater social distance implies less guilt per unit consumed, and a stronger windfall function implies more windfall benefit per unit consumed. Together, these assumptions imply $\partial x_o^* / \partial s > 0$: optimal other-earned consumption is higher at greater social distance (derivation in Appendix A).

At sufficiently high social distance, windfall benefit dominates guilt cost and the net term $g_x - w_x$ is unambiguously negative, so $x_o^* > x_s^*$: the consumer is willing to spend more when the money was earned by a distant other than when it was earned by themselves. This prediction reflects the specific environment of the model, one in which the joint account holds opportunity cost constant across scenarios by the joint account assumption. In most real-world settings where a spender has implicit or explicit permission to spend another person's money, property rights are not the binding constraint; rather, the effective opportunity cost of spending other-earned funds is typically lower, because unspent funds revert to the earner rather than being split equally. That additional force would push further toward higher spending with other-earned money, beyond what guilt and windfall alone predict. The experiments described in Section 4 provide upper-bound estimates of the behavioral effect of other-earned money on spending behavior. By equalizing opportunity costs across conditions, they isolate the net psychological term but likely understate how much more freely people would spend other-earned money in the field, where the effective opportunity cost is lower.

3.3. Empirical Implications

Willingness to pay is the behavioral counterpart to pain of paying in this framework. Pain of paying raises the perceived cost of consumption and thereby reduces the maximum price a consumer is willing to pay for a given good. If other-earned spending generates higher pain of paying than self-earned spending, the model predicts correspondingly lower willingness to pay with other-earned money — and vice versa if pain of paying is lower. In the ideal case the relationship is one-to-one: each unit increase in pain of paying produces an equal reduction in willingness to pay. In practice the mapping may be weaker, and the two outcomes need not move together. A divergence between the two outcomes (pain of paying differs while willingness to pay does not differ) indicates that psychological costs are present but do not fully translate into behavioral differences.

At low social distance, the model provides no directional prediction on willingness to pay,

as the sign of $g_x - w_x$ is ambiguous. The pain of paying may still differ across self- and other-earned spending, even if economic behavior remains the same. Even when behavior is unchanged, pain of paying can differ across conditions because the windfall and guilt terms shift the affective cost of spending above or below the opportunity cost alone:

$$p_o = c(x) + g(x, s) - w(x, s) \neq c(x) = p_s . \quad (6)$$

Holding opportunity cost constant by design, the couples experiment identifies the net $g - w$ term as the sole channel through which money source affects spending. This motivates the pain-of-paying elicitation as a complementary outcome capable of detecting psychological effects that may not manifest in purchasing behavior.

At high social distance — the case of randomly matched strangers — the comparative static in Section 3.2 delivers a clear directional prediction. Guilt costs are minimal and windfall benefits are substantial at high s , and optimal consumption with other-earned money exceeds optimal consumption with self-earned money, $x_o^* > x_s^*$. A parallel implication holds for pain of paying. At higher values of s , $g(x, s)$ is smaller and $w(x, s)$ is larger, placing the pain of paying with other-earned money below the self-earned level once s is sufficiently large. The inclusion of the stranger extension tests this prediction directly. Taken together, the two experiments span the social distance gradient from its lower bound (romantic partners) to the upper bound achievable in a laboratory setting (randomly matched strangers), providing a joint test of the model’s core comparative static.

4. Experimental Design: Couples Study

The couples experiment tests how money source affects willingness to pay and pain of paying when individuals spend from a shared joint account; Section 4.3 describes a planned extension using randomly matched strangers. In the couples experiment, one member of each couple acts as the spender across two rounds. The two rounds differ by treatment: in the self-earned treatment, the spender purchases an item using money they earned themselves; in the partner-earned treatment, they purchase the same item using money earned by their partner. Couples were randomly assigned to one of two order conditions (self-earned first or partner-earned first), determining which treatment the spender experienced in Round 1. Following both rounds, all participants rated their anticipated pain of paying for hypothetical purchases at a range of price points under both treatments and answered a series of post-experiment questions.

The experiment was approved by the University of Wyoming Institutional Review Board (IRB-2024-68) and analyses were pre-registered with the AEA RCT Registry (AEARCTR-0015737). Sessions were conducted at the University of Wyoming Teton Lab using oTree (Chen et al., 2016). Data collection spanned May–June 2025 and September–November 2025, with final sessions planned for April–May 2026.

Section 4.1 presents participant characteristics and recruitment; Section 4.2 describes the experimental procedures in full. Section 4.3 describes a planned extension using randomly matched strangers. Full experimental instructions are provided in Appendix B.

4.1. Participants and Recruitment

Participants were eligible if they were in a romantic relationship (dating, engaged, or married), could attend the lab session together with their partner, and were both at least 18 years old. No restrictions were placed on relationship length. Since couple participation was required, recruitment drew broadly from the University of Wyoming community and the surrounding Laramie population. Flyers were posted on campus and throughout the local community; recruitment messages were sent through email lists to students, staff, and faculty at the University of Wyoming and advertised on local Facebook groups.

As of 6th October 2026, 100 couples (71% of the target sample) had completed the experiment; sample characteristics are reported in Table 1.² The average participant age was approximately 24 years; 51% were female. Relationship length was categorized as short (< 3 years), medium (3–10 years), or long (> 10 years); most couples fell into the short (55%) or medium (39%) categories. Approximately half of couples cohabited at the time of the experiment; 25% of the full sample were married. Six couples (6%) were same-sex couples. Participants received a \$15 show-up fee, half of the remaining joint account balance from the randomly selected payment round, and \$1 if their ranking prediction met the accuracy threshold, for average total earnings of \$24.69. Sessions lasted approximately 30 minutes.

2. In total, 112 couples completed sessions; data from 12 couples across two sessions were lost due to software errors, leaving a usable sample of 100 couples.

Table 1: Sample Characteristics

Variable	Mean (SD) or %	N
<i>Demographics (N = 200 individuals)</i>		
Age (years)	23.9 (8.6)	200
Female (%)	50.5	101
<i>Relationship Characteristics (N = 100 couples)</i>		
Relationship duration:		
Short (< 3 years)	55.0%	55
Medium (3–10 years)	39.0%	39
Long (> 10 years)	6.0%	6
Cohabiting (%)	53.0	53
Married (%)	25.0	25
Same-sex couple (%)	6.0	6

Notes: Demographics reported at the individual level ($N = 200$ individuals from 100 couples). Relationship characteristics reported at the couple level ($N = 100$ couples).

4.2. Experimental Procedures

The experiment proceeded in six stages: welcome and consent, item preference ranking, a real-effort earnings task, a willingness-to-pay elicitation, a pain-of-paying elicitation, and a post-experiment questionnaire.

Welcome and consent. Participants arrived at the lab together and were seated at separate computer stations to ensure independent responses. After providing informed consent, each couple was randomly assigned to an order: self-earned first ($N = 44$ couples) or partner-earned first ($N = 56$ couples), determining which treatment the spender experienced in Round 1. Within each couple, one partner was randomly assigned the spender role for the duration of the experiment; the other partner served as the earner in the round in which the spender spent self-earned money, and as the passive participant in the other round.

Item preference ranking. Both partners then ranked a list of ten consumer goods, each valued at approximately \$30, according to their own preferences, from most to least preferred. After submitting their own preferences, each participant predicted their partner's rankings of the same items. Predictions were incentivized: participants earned an additional \$1 if at least five of their predicted rankings matched their partner's true rankings exactly. These rankings determined which item each participant was offered in the willingness-to-pay elicitation. An algorithm selected the item with the largest gap between the participant's own preference rank and their predicted rank for their partner: the item the participant ranked highly for themselves but expected their partner to rank low. This difference in ranking ensured the spender was bidding on an item they genuinely wanted for themselves rather than an item they might purchase with their partner in mind. Without this selection rule, a spender who happened to be offered an item they disliked but believed their partner preferred might report higher willingness to pay out of consideration for their partner, confounding the treatment comparison.

Ten items were chosen to provide sufficient variety in preferences without inducing decision fatigue; the items spanned categories stereotypically associated with male or female preferences, as well as gender-neutral options, to ensure meaningful preference divergence within couples. Offering multiple items served a second purpose: by selecting the item the spender ranked most highly for themselves relative to their predicted partner ranking, the algorithm maximized the spender's own valuation for the offered item. Offering a single fixed item to all participants would have introduced substantial variation in base willingness to pay driven by idiosyncratic item preferences rather than money source, obscuring the treatment contrast of interest. Ties were broken by selecting the item the participant ranked highest among those tied. A description of the item selection process and the complete list of items displayed can be found in [Appendix C](#).

Real-effort earnings task. In each round, one member of the couple (the earner) completed a real-effort earnings task while the other waited. A real-effort task was used to ensure that earnings felt genuinely earned rather than a windfall subsidy; this distinction is essential for the treatment contrast, because the goal is to compare spending of self- versus partner-earned money, not spending of earned versus unearned money. The task involved counting the number of 1s in a series of 5×5 matrices of 1s and 0s. The matrix task was selected in part because it does not systematically advantage any gender (Abeler et al., 2011; Choi et al., 2018; Kachelmeir et al., 2016). Gender-neutrality in the earnings task

was important consideration given that the experiment targets heterosexual couples and a task perceived as easier by one gender could confound treatment differences through differential effort salience. Participants completed three practice matrices before Round 1; during Rounds 1 and 2, they had five minutes to complete five paid matrices, earning \$4 per correctly completed matrix for a maximum of \$20. All couples completed all five matrices; if participants had not finished within the time limit, their earnings would be reduced proportionally.

Treatment assignment determined who completed the earnings task in Round 1: in the self-earned first order, the spender completed the task; in the partner-earned first order, the spender's partner did. In Round 2, roles reversed, so that the spender experienced both treatments across the two rounds.

Earnings from each round were deposited into a joint account shared equally by both partners, giving each partner equal property rights over the funds regardless of who completed the task. As discussed in Section 3, the joint account serves two purposes: eliminating the property rights confound and holding opportunity cost constant across treatments. By splitting unspent funds equally regardless of treatment, the joint account ensures that a dollar spent is a dollar not split in both conditions, isolating the net psychological effect of money source from any difference in opportunity cost.

Willingness-to-pay elicitation. Following the earnings task, the spender stated their maximum willingness to pay for the item selected by the algorithm. Willingness to pay was elicited using a Becker–DeGroot–Marschak (BDM) auction (Becker et al., 1964), which provides incentive-compatible willingness to pay estimates without requiring strategic interaction between participants. In a BDM auction, participants state their maximum willingness to pay for an item, and a random price is drawn. If the stated willingness to pay meets or exceeds the random price, the participant purchases the item at the random price, with the cost deducted from the joint account; otherwise, no transaction occurs.

The BDM auction is incentive-compatible and the dominant strategy is truthful revelation. Overstating willingness to pay risks paying more than the item is worth (if the random price falls between true value and stated willingness to pay); understating willingness to pay risks forgoing a desirable purchase (if the random price falls between stated and true willingness to pay). Participants were informed that prices were randomly drawn, without

specifying the specific distribution.³ Participants answered comprehension questions before proceeding; correct answers were displayed regardless of their response to reinforce understanding.

Items were valued at approximately \$30, but participants could state a willingness to pay of at most \$20 — the joint account balance. This allowed participants to obtain items at a meaningful discount relative to market value, but only by spending from the joint account. In the partner-earned treatment, a participant who valued an item at \$30 and states a willingness to pay of \$20 was virtually guaranteed to receive it, but only by spending the full joint account balance, half of which (\$10) belonged to their partner. This design ensures that spending decisions involve meaningful trade-offs regardless of money source. Each round used an independent joint account; at the experiment's conclusion, one round was randomly selected for payment, and only that round's purchase decision was binding.

Pain-of-paying elicitation. After both rounds concluded but before the payment round was randomly selected, all participants — both spenders and non-spenders — rated their anticipated pain of paying for the item selected by the algorithm. Pain of paying was measured using the smiley Likert scale shown in Figure 1, which captures affective responses to pain on a seven-point ordinal scale. Participants rated their anticipated pain of paying at nine price points (\$0, \$2, \$4, \$6, \$8, \$10, \$12, \$16, and \$20) under both the self-earned and partner-earned conditions, for a total of 18 ratings.

Price points were selected at \$2 increments up to \$12, then at \$16 and \$20. Eliciting ratings at every dollar value would have required 40 decisions per participant; the \$2 increment reduced this to a manageable number while preserving coverage across the full range. The upper end of the scale was thinned on the expectation that pain ratings would cluster heavily toward the maximum at high price points, limiting variation there. The \$0 price point was included as an anchor and a design check: under either condition, participants would be expected to report no pain of paying at zero cost.

Eliciting pain ratings after both rounds but before the payment round was selected provided measurements across the full price range and prevented outcome knowledge from contaminating pain ratings. The elicitation captures anticipated rather than experienced pain at the moment of decision, a tradeoff against those advantages.

3. Prices were drawn from $N(17, 0.5^2)$ rather than a more typical $U[0, 20]$. The high-mean distribution kept purchase rates low while preserving incentive compatibility, reducing the logistical burden of purchasing and shipping items to the lab.

For the primary analysis, we focus on the \$10 price point, which represents each participant's equal share of the \$20 endowment. As an exploratory robustness check, we also examined the \$2 price point, which was the median willingness to pay among spenders who stated different bids across treatments.

Post-experiment questionnaire. After the pain-of-paying elicitation, participants completed a post-experiment questionnaire covering relationship characteristics and demographics. Key relationship characteristics included relationship length, cohabitation status, marital status, financial integration, relationship satisfaction, and household composition. Spenders reported their feelings about spending from the joint account, including guilt, sense of entitlement to self- versus partner-earned funds, and comfort with potential partner disapproval. Non-spenders reported analogous measures from the observer's perspective: expected guilt in their partner, perceived partner entitlement to the funds, and anticipated displeasure at joint account spending. The spendthrift–tightwad scale from Rick et al. (2008) was also administered to examine whether individual differences in spending orientation moderated willingness to pay. Demographic measures included age, gender, education, student status, and household income. One round was then randomly selected for payment.

4.3. Extension: Experiment with Strangers

The stranger extension applies the same design as the couples experiment, differing only in participant matching. Rather than attending the lab with their romantic partner, participants are paired with a randomly matched stranger. All other procedures, including the two-round structure, joint account, real-effort earnings task, and BDM auction, are maintained to ensure comparability across experiments.

As in the couples design, strangers' earnings are placed in a joint account with equal property rights. The joint account is maintained in the stranger extension to hold property rights and opportunity cost constant across experiments, ensuring that observed differences in willingness to pay or pain of paying reflect social distance alone rather than differences in perceived ownership or effective budget. The joint account is not a natural feature of stranger interactions, but removing it would confound social distance with opportunity cost.

Data collection for the stranger extension will begin upon completion of the couples experiment. The design is finalized and the oTree implementation is being adapted from

the couples experiment. IRB approval and AEA RCT Registry pre-registration are pending. We target 140 stranger pairs (280 participants) to match the couples sample.

5. Results

Money source did not significantly affect willingness to pay but increased the pain of paying. Spenders bid slightly more with self-earned money than with partner-earned money, but the difference is small and statistically indistinguishable from zero: mean willingness to pay was \$4.95 ($SD = \5.60) in the self-earned condition and \$4.47 ($SD = \5.54) in the partner-earned condition, a mean difference of \$0.48 ($SD = \2.95). Individual-level differences are highly concentrated at zero (Figure 2).

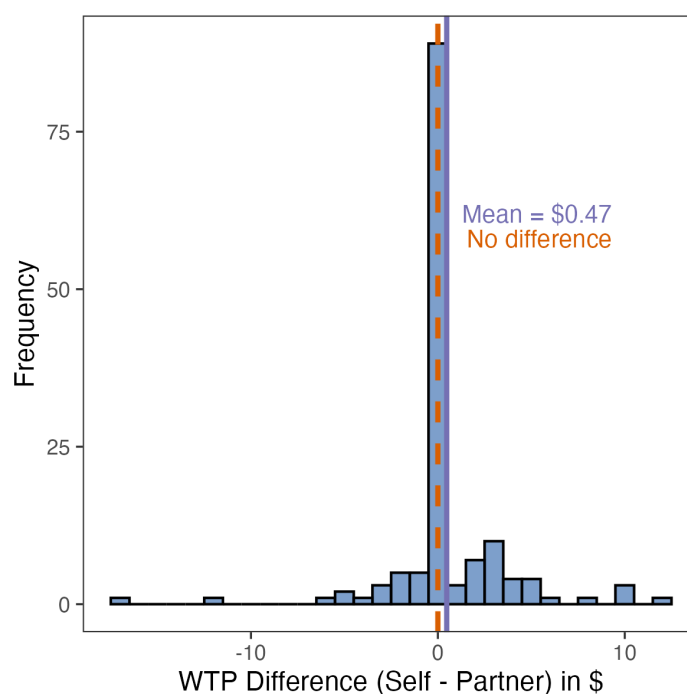


Figure 2: Distribution of within-subject willingness to pay differences (Self–Partner) across 100 spenders. The vertical blue line marks the sample mean (\$0.48); the dashed orange line marks zero. Sixty-one percent of spenders recorded a difference of exactly \$0.

All three tests of the within-subject difference (a paired t -test, Wilcoxon signed-rank test, and sign test) fail to reject the null (Table 2). Cohen’s $d = 0.087$ is negligible by conventional

benchmarks, confirming the practical insignificance of the difference. The \$0.48 average difference is small relative to the standard deviation of individual differences, and the 95% confidence interval on the mean difference (−\$0.10 to \$1.07) includes zero.

Table 2: Main Treatment Effects: Willingness to Pay and Pain of Paying

Outcome	Condition	<i>N</i>	Mean	SD	Difference	<i>p</i> -value	Cohen's <i>d</i>
<i>Panel A: Willingness to Pay (\$)</i>							
	Self-earned	100	4.95	5.60	+0.48	0.071	0.087
	Partner-earned	100	4.47	5.54			
<i>Panel B: Pain of Paying (1–7 scale, \$10 price point)</i>							
	Self-earned	100	4.15	2.20	−1.07	<0.001	−0.518
	Partner-earned	100	5.22	1.86			

Notes: $N = 100$ spenders from 100 couples. Difference = Self – Partner. p -values from Wilcoxon signed-rank tests (primary); paired t -test results are consistent throughout. Cohen's d computed on paired differences. Both parametric and non-parametric tests reported given methodological debate on Likert-scale data. Results are preliminary.

Treatment order did not affect within-subject comparisons. Spenders in the self-earned first order recorded a mean willingness to pay difference of \$0.64, compared to \$0.36 among spenders in the partner-earned first order; a Welch t -test for equality of these means yields $t(97.2) = -0.48$, $p = 0.632$.

As a robustness check, Round 1 data provide a clean between-subjects comparison: spenders were randomly assigned to conditions in the first round before experiencing both treatments. Among spenders randomly assigned to the self-earned condition in Round 1 ($n = 44$), mean willingness to pay was \$5.67 ($SD = \5.64); among those assigned to the partner-earned condition ($n = 56$), mean willingness to pay was \$4.02 ($SD = \5.58). A Wilcoxon rank-sum test yields $W = 1,466$, $p = 0.094$, and Cohen's $d = 0.294$ (Figure 10, Appendix D). The direction is consistent with the within-subject result; the between-subjects effect is significant at the 10% level ($p = 0.094$), though neither design reaches the conventional 5% threshold.

5.1. Response Patterns

Individual spenders responded to the treatment heterogeneously, even though the average effect is null. Four mutually exclusive response patterns emerge: 32% bid zero in both conditions, 29% bid identical positive amounts, 24% bid higher with self-earned money, and 15% bid higher with partner-earned money (Table 3). Among the 61 spenders who bid identically across conditions, money source produced no discernible behavioral effect. Among the 39 spenders whose bids differed, 24 (62%) bid higher with self-earned money and 15 (38%) bid higher with partner-earned money. Where a treatment effect appears at all, the direction is consistent with higher willingness to pay in the self-earned condition (Figure 3).

Table 3: Willingness to Pay Response Patterns Across Conditions

Pattern	<i>N</i>	%	Mean WTP Self (\$)	Mean WTP Partner (\$)	Mean Difference (\$)
Both Zero	32	32	—	—	—
Identical Positive	29	29	6.99	6.99	—
Higher with Self	24	24	8.28	4.23	+4.05
Higher with Partner	15	15	6.23	9.50	-3.27
<i>Full sample</i>	<i>100</i>	<i>100</i>	<i>4.95</i>	<i>4.47</i>	<i>+0.48</i>

Notes: *N* = 100 spenders from 100 couples. Response patterns defined by bidding behavior across conditions. *Both Zero*: bid \$0 in both conditions. *Identical Positive*: stated the same positive amount in both conditions. *Higher with Self*: bid more in the self-earned condition. *Higher with Partner*: bid more in the partner-earned condition

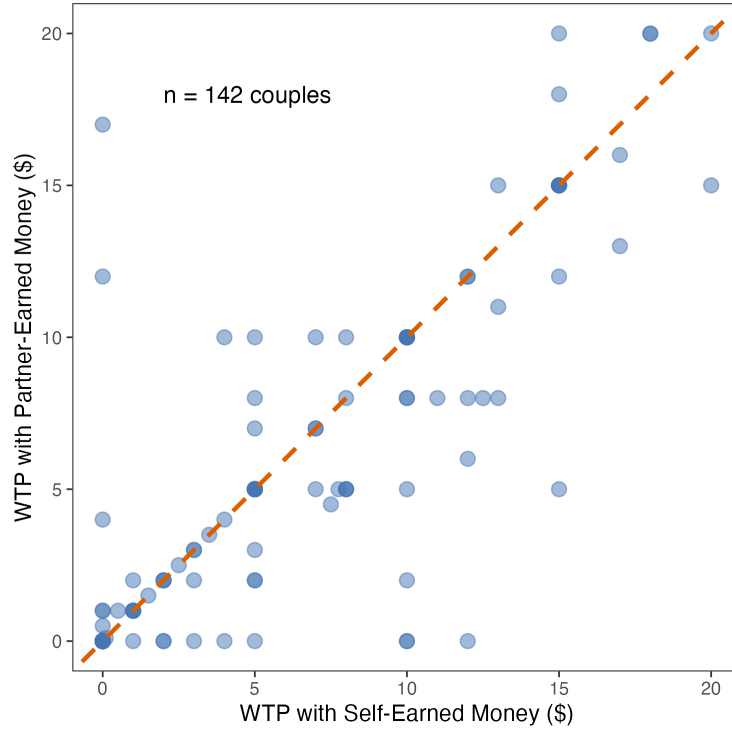


Figure 3: Willingness to pay in the self-earned condition plotted against willingness to pay in the partner-earned condition for each of the 100 spenders. Points on the dashed 45-degree line indicate identical bids across conditions; points above the line indicate higher willingness to pay with partner-earned money.

Among the 39 spenders who varied their bids, a Wilcoxon signed-rank test yielded $p = 0.071$ and Cohen's $d = 0.217$, a small-to-medium effect. Mean willingness to pay in this subset was \$7.49 ($SD = \5.52) in the self-earned condition and \$6.26 ($SD = \5.86) in the partner-earned condition, a mean difference of \$1.24 ($SD = \4.65; Figure 11, Appendix D). This post-hoc, exploratory analysis characterizes the concentration of the treatment effect rather than testing the primary hypothesis.

5.2. Pain of Paying

In contrast to the null willingness-to-pay result, spenders reported consistently higher pain of paying when spending their partner's money than their own (Table 4). Partner-earned pain exceeds self-earned pain at every price point; the gap peaks at the \$10 price point, each partner's equal share of the joint account endowment, where mean ratings were 4.15 ($SD = 2.20$) in the self-earned condition and 5.22 ($SD = 1.86$) in the partner-earned

condition. Partner-earned pain ratings are clustered heavily at the high end of the scale, while self-earned ratings are more evenly distributed across the range (Figure 4).

Table 4: Pain of Paying Ratings For Spenders Across Price Points

Price (\$)	Mean Self	SD Self	Mean Partner	SD Partner	Mean Diff
0	1.07	0.61	1.12	0.84	-0.05
2	1.70	1.52	2.31	1.84	-0.61
4	2.19	1.64	3.07	2.01	-0.88
6	2.84	1.89	3.83	2.07	-0.99
8	3.39	1.99	4.45	2.03	-1.06
10	4.15	2.20	5.22	1.86	-1.07
12	4.99	1.92	5.84	1.69	-0.85
16	5.67	1.58	6.30	1.34	-0.63
20	6.06	1.44	6.47	1.27	-0.41

Notes: $N = 100$ spenders from 100 couples. Mean pain ratings on the smiley Likert scale (1 = no pain, 7 = maximum pain) at each price point under the self-earned and partner-earned conditions. Mean Diff = Mean Self – Mean Partner. The \$10 price point is the primary outcome; the \$2 price point is used in the robustness check in the text.

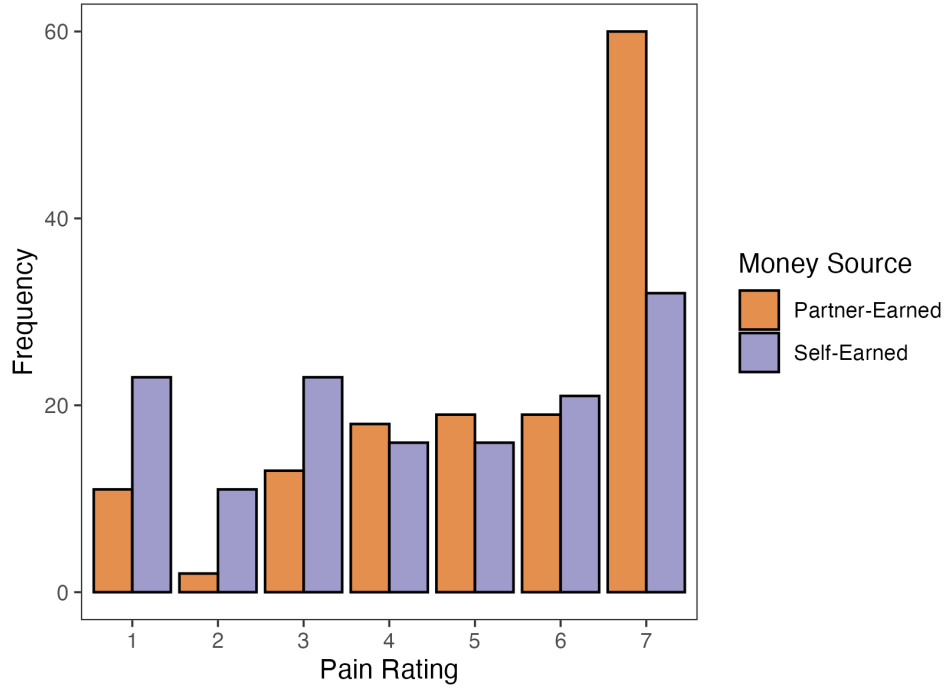


Figure 4: Pain-of-paying distributions at the \$10 price point for the self-earned and partner-earned conditions ($N = 100$ spenders). The scale runs from 1 (no pain) to 7 (maximum pain).

Both a Wilcoxon signed-rank test ($V = 98.5$, $p < 0.001$) and a paired t -test ($t(99) = -6.14$, $p < 0.001$) reject the null of no difference. Both parametric and non-parametric tests are reported given ongoing methodological debate about the appropriate treatment of Likert-scale data. Individual differences are strongly asymmetric: 48% of spenders reported higher pain with partner-earned money, 47% reported equal pain, and only 5% reported lower pain. The 48-to-5 asymmetry indicates the effect is broadly distributed across participants rather than driven by a small number of outliers.

As a post-hoc robustness check, pain ratings at the \$2 price point — the median willingness to pay among spenders who stated different bids across conditions — yielded qualitatively identical results ($p < 0.001$ for both Wilcoxon and paired t -test; mean difference = $-\$0.61$). This price point was not pre-specified; it was chosen because it represents the margin at which behaviorally responsive spenders were roughly indifferent.

The primary analysis focuses on the 100 spenders, who faced actual spending decisions and whose pain ratings directly reflect that context. As a robustness check, including

non-spenders — who rated hypothetical pain for their partner’s item without having made a bid — yields qualitatively consistent results: the null of no difference is rejected across both parametric and non-parametric tests ($p = 0.018$ and $p = 0.005$ respectively), with mean ratings of 3.55 ($SD = 2.03$) in the self-earned condition and 3.90 ($SD = 2.22$) in the partner-earned condition (Cohen’s $d = 0.161$; Table 6, Appendix D). The effect is smaller in the full sample, consistent with the expectation that non-spenders answered more abstractly and with less psychological engagement than those who actually faced the spending decision.

The correlation between individual willingness to pay differences and pain-of-paying differences is positive but small and not statistically significant: Spearman’s $\rho = 0.144$, $p = 0.154$; Pearson’s $r = 0.013$, $p = 0.901$ (Figure 12, Appendix D). Spenders who adjusted their willingness to pay across conditions did not systematically report larger pain differences, and vice versa. The two measures appear to capture distinct dimensions of the spending experience: pain of paying is sensitive to money source even when stated willingness to pay is not, suggesting the psychological cost of spending partner-earned money is real but does not consistently translate into behavioral differences.

5.3. Heterogeneity Analysis

The following analyses are exploratory: subgroup samples are small, the tests are underpowered, and no corrections for multiple comparisons have been applied. The results are reported to characterize patterns in the preliminary data and motivate future inquiry, not to draw firm conclusions.

Cohabitation. Non-cohabiting couples ($n = 45$) show a mean willingness to pay difference of \$0.78, and a within-group Wilcoxon signed-rank test yields $p = 0.072$, marginally significant. Cohabiting couples ($n = 55$) show a mean willingness to pay difference of \$0.24 and a within-group Wilcoxon $p = 0.454$. A between-group test of whether willingness to pay differences vary with cohabitation status finds no significant difference (Welch t -test $p = 0.356$; Wilcoxon rank-sum $p = 0.691$; Cohen’s $d = 0.186$). The pattern is nonetheless suggestive: non-cohabiting couples, who tend to be less financially integrated, show a larger and marginally significant within-group treatment effect, while cohabiting couples show no discernible response. This is consistent with the idea that greater financial integration between partners attenuates the psychological distinction between self- and partner-earned funds, though the between-group test does not reach conventional

significance levels at this sample size. Cohabitation subgroup distributions are provided in Figure 13 (Appendix D).

Gender. Male spenders ($n = 47$) recorded a mean willingness to pay difference of \$0.19; female spenders ($n = 49$) recorded a mean difference of \$0.62. Non-binary spenders ($n = 4$) were excluded from the subgroup analysis. Neither within-group Wilcoxon test approaches significance (male: $p = 0.393$; female: $p = 0.124$), and the between-group test finds no significant gender difference (Welch t -test $p = 0.440$; Cohen's $d = -0.159$). Gender subgroup distributions are shown in Figure 14 (Appendix D).

Relationship duration. Mean willingness to pay differences across relationship duration categories are \$0.41 (short, < 3 years; $n = 56$), \$0.59 (medium, 3–10 years; $n = 37$), and \$0.50 (long, > 10 years; $n = 7$). A Kruskal-Wallis test finds no significant variation across groups ($\chi^2 = 0.071$, $df = 2$, $p = 0.965$). The long-duration group is too small for meaningful inference. Duration subgroup distributions are shown in Figure 15 (Appendix D).

5.4. Power and Sample Size

The observed willingness to pay effect was small: Cohen's $d = 0.087$ on the paired difference. Post-hoc power at 100 couples is 0.37, well below the conventional threshold of 0.80. At the target sample of 140 couples, power for an effect of this magnitude would reach approximately 0.43, still substantially below threshold. The null willingness to pay result should therefore be interpreted cautiously: the study as currently designed cannot rule out a small but potentially meaningful difference in willingness to pay. The minimum detectable effect at 80% power is $d = 0.283$ (\$0.83 in dollar terms) at $n = 100$ and $d = 0.238$ (\$0.70) at $n = 140$; the observed effect falls well below both thresholds.

Power curves across a range of effect sizes are provided in Figure 16. At $d = 0.2$, power reaches 0.65 at the target sample of 140 couples; at $d = 0.3$, power reaches 0.94. Completing the target sample will improve precision but will not be sufficient to detect an effect as small as the one observed here with high confidence. By contrast, the pain-of-paying result ($d = 0.518$) is well above the minimum detectable effect at both sample sizes, and power is not a concern for that outcome.

The stranger extension is designed to test the model's comparative static prediction — that willingness to pay is higher with stranger-earned money than with self-earned money — at a social distance where the directional prediction is unambiguous and the expected

effect size may be substantially larger.

6. Discussion

Money source shapes the psychological experience of spending even when it leaves purchasing behavior unchanged. This finding holds at the low end of the social distance spectrum, among romantic couples. Spenders report consistently higher pain of paying when spending their partner's money than their own — a medium-sized effect that is robust across parametric and non-parametric tests and broadly distributed across participants. Willingness to pay, by contrast, does not differ significantly across conditions on average.

Three interpretations of the null willingness-to-pay result are consistent with the data. First, the null may reflect a genuine psychological equilibrium at low social distance: windfall benefits from spending unearned funds and guilt costs from potentially disadvantaging a close partner may be roughly equal in magnitude among romantic couples, producing no net effect on purchasing behavior. This interpretation is consistent with the theoretical framework, which makes no directional prediction at low social distance precisely because both forces are active and neither dominates in general.

Second, the within-subject design may have created consistency pressure that compressed treatment responses toward zero. Participants who completed both conditions in the same session may have anchored their second bid to their first, compressing individual-level differences toward zero. The between-subjects robustness check using Round 1 data only provides some reassurance: the direction of the effect is consistent with the within-subject result, and the between-subjects comparison yields a larger effect significant at the 10% level ($p = 0.094$). This consistency between designs is difficult to explain if anchoring or consistency pressure alone were driving the null.

Third, the study may be underpowered to detect a small but real effect. The observed effect size (Cohen's $d = 0.087$) falls well below the minimum detectable effect at both the current and target sample sizes, and post-hoc power at $N = 100$ is 0.37. The data are consistent with a true null but cannot rule out a small effect below the design's minimum detectable threshold. The stranger extension tests the model's comparative static at high social distance, where the directional prediction is unambiguous; its results will help distinguish among these three interpretations.

The composition of the null result further complicates its interpretation. The average null is driven largely by the 61% of spenders who bid identically across conditions, including the 32% who bid \$0 in both. Among the 39 spenders whose bids differed, the pattern is directionally consistent with higher willingness to pay under the self-earned condition (24 of 39, or 62%, bid more with self-earned money, mean difference \$1.24 ($SD = \4.65)), and a Wilcoxon signed-rank test on this subset yields $p = 0.071$. If anything, these results suggest that spenders who responded to the treatment tended to spend *less* with partner-earned money, the opposite of the Friedman conjecture, though the effect does not reach conventional significance even in this subset.

The pain-of-paying result admits a more direct interpretation. Spenders feel meaningfully worse about spending their partner's money than their own, even when their stated willingness to pay is the same. This divergence suggests that the psychological cost of spending partner-earned money is real but does not consistently translate into purchasing behavior. The asymmetry in individual-level pain differences suggests the effect is broadly distributed rather than driven by outliers. The result also replicates and extends Enriquez et al. (2019), who found higher pain of paying with partner-earned money in a hypothetical survey setting; we found the same pattern in an incentivized laboratory experiment with real purchasing decisions.

The couples experiment establishes a baseline at the low end of the social distance spectrum. At this end, the theory is deliberately agnostic about direction, and the data reflect that ambiguity: willingness to pay was statistically indistinguishable across conditions while pain of paying diverged sharply. The stranger extension tests the other end of the spectrum achievable in a laboratory setting. At high social distance, guilt costs are minimal and windfall benefits dominate, and the theory predicts unambiguously that willingness to pay will be higher with other-earned money. The two experiments together span the social distance gradient and provide a joint test of the model's comparative static. If the stranger extension finds the predicted positive effect on willingness to pay, it will strengthen the case that the null among romantic couples reflects genuine offsetting forces rather than a design artifact or insufficient power. If the null persists at high social distance, that would cast doubt on the windfall-guilt framework and motivate a reassessment of whether social distance is the right organizing variable.

These results speak to Friedman and Friedman's (1990) conjecture that people spend other people's money more wastefully than their own — a claim that, to our knowledge, has not

been directly tested in an incentivized laboratory setting. The evidence from the couples experiment points against this conjecture, at least at low social distance. Spenders report higher psychological costs when spending partner-earned money, and the directional willingness to pay effect, where it appears at all, runs opposite to the wasteful-spending prediction: among the 39 spenders who bid differently across conditions, 62% bid *more* with self-earned money. Holding opportunity cost constant by design, the experiment isolates the net psychological effect of money source and finds no evidence of the uninhibited spending Friedman and Friedman (1990) anticipated, even when the opportunity cost channel is closed off. Whether the conjecture holds at greater social distance, where guilt diminishes and the effective opportunity cost of spending other-earned money is typically lower, is precisely what the stranger extension is designed to address.

Three design features warrant careful interpretation. The within-subject design improved statistical power but may have introduced consistency pressure that compressed individual-level willingness-to-pay differences; the between-subjects robustness check partially addresses this concern. The pain-of-paying elicitation captures anticipated rather than experienced pain at the moment of decision, a tradeoff against the advantages of full price-range coverage and pre-outcome elicitation timing. For the six same-sex couples in the sample, the item selection algorithm, which was designed around gender-differentiated preferences, may not have optimally reflected preference divergence between partners.

People spend other people's money constantly — in households, firms, and governments. Friedman and Friedman's (1990) conjecture that people spend other people's money more wastefully than their own has now been put to an initial test. The findings here complicate that picture. Among romantic couples, where social distance is minimal and fairness norms are strong, money source leaves purchasing behavior unchanged while raising the psychological cost of spending. Whether that behavioral null gives way to the predicted increase in spending as social distance grows — from partners to strangers to colleagues to bureaucrats — is precisely what the stranger extension is designed to test. If the theory is right, the guilt that disciplines spending in close relationships fades with distance, and the Friedman conjecture holds, but only at the far end of the social distance spectrum, and only when opportunity costs are not held constant.

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Appendices

A. Theoretical Appendix

This appendix presents the full set of assumptions underlying the theoretical framework in Section 3 and derives the comparative static in Section 3.2.

A.1. Functional Form Assumptions

The enjoyment function $u(x)$ and opportunity cost function $c(x)$ satisfy standard regularity conditions: $u' > 0$, $u'' < 0$, $c' > 0$, and $c'' > 0$. Enjoyment is increasing and concave, reflecting diminishing marginal utility from consumption, and opportunity cost is increasing and convex, reflecting an accelerating marginal cost of foregone alternatives.

The windfall function $w(x, s)$ satisfies four assumptions beyond those stated in the main text. First, windfall benefit is concave in consumption ($w_{xx} < 0$): the marginal psychological gain from spending unearned funds diminishes as consumption increases, so the first unit spent generates more windfall benefit than subsequent units. Second, windfall benefit is concave in social distance ($w_{ss} < 0$): the marginal increase in windfall benefit from greater social distance diminishes as the consumer becomes more distant from the earner, with the windfall effect plateauing at very high social distance. Third, the marginal windfall benefit of consumption is increasing in social distance ($w_{xs} > 0$): at greater social distance, each additional unit of consumption generates more windfall benefit because the consumer is less attentive to the earner's interests and more focused on the gain from spending unearned funds. Fourth, zero consumption generates zero windfall benefit ($w(0, s) = 0$): the windfall effect is activated only through the act of spending.

The guilt function $g(x, s)$ satisfies four analogous assumptions. First, guilt cost is convex in consumption ($g_{xx} > 0$): the marginal psychological cost of spending unearned funds accelerates as consumption increases, reflecting heightened fairness concerns and anticipated disapproval at higher spending levels. Second, guilt cost is convex in social distance ($g_{ss} > 0$): the marginal decrease in guilt from greater social distance diminishes as the consumer becomes more distant from the earner, with guilt approaching but never reaching zero at very high social distance. Third, the marginal guilt cost of consumption is decreasing in social distance ($g_{xs} < 0$): at greater social distance, each additional unit of consumption generates less guilt because the consumer feels less responsible for the earner's wellbeing. Fourth, zero consumption generates zero guilt cost ($g(0, s) = 0$): guilt

arises only through the act of spending another person's funds.

A.2. Derivation of the Comparative Static

The comparative static $\partial x^*/\partial s$ is derived by applying the implicit function theorem to the first-order condition for the other-earned problem. At the optimum x^* , the first-order condition holds with equality for all values of s , so define

$$F(x^*, s) \equiv u'(x^*) - c'(x^*) + w_x(x^*, s) - g_x(x^*, s) = 0.$$

Since $F = 0$ must hold for any value of s , a perturbation in s requires a corresponding adjustment in x^* to maintain the equality. Totally differentiating F with respect to s via the chain rule yields

$$\frac{\partial F}{\partial x^*} \cdot \frac{\partial x^*}{\partial s} + \frac{\partial F}{\partial s} = 0,$$

which captures two channels through which s affects F : directly through the windfall and guilt functions, and indirectly through its effect on the optimal consumption level x^* . Solving for the comparative static,

$$\frac{\partial x^*}{\partial s} = -\frac{\partial F/\partial s}{\partial F/\partial x^*}.$$

Differentiating F with respect to s , holding x^* fixed, gives the numerator. The terms $u'(x^*)$ and $c'(x^*)$ do not depend on s directly and therefore drop out, leaving

$$\frac{\partial F}{\partial s} = w_{xs}(x^*, s) - g_{xs}(x^*, s).$$

The assumptions $w_{xs} > 0$ and $g_{xs} < 0$ imply that both terms are positive, so $\partial F/\partial s > 0$.

Differentiating F with respect to x^* , holding s fixed, gives the denominator,

$$\frac{\partial F}{\partial x^*} = u''(x^*) - c''(x^*) + w_{xx}(x^*, s) - g_{xx}(x^*, s).$$

The assumptions $u'' < 0$, $c'' > 0$, $w_{xx} < 0$, and $g_{xx} > 0$ imply that all three terms are negative, so $\partial F/\partial x^* < 0$. This condition also corresponds to the second-order condition for an interior maximum of U : the first-order condition is locally decreasing in x^* , reflecting both the standard concavity of u and c and the curvature assumptions on w and g together, confirming that x^* is a maximum rather than a minimum.

Combining the signed numerator and denominator,

$$\frac{\partial x^*}{\partial s} = -\frac{(+)}{(-)} > 0,$$

so optimal consumption with other-earned money is higher at greater social distance between consumer and earner. This confirms the result stated in Section 3.2: optimal other-earned consumption is increasing in social distance.

B. Experimental Instructions

This appendix reproduces the full text of the experimental instructions shown to participants during each session. Instructions were implemented in oTree (Chen et al., 2016) and displayed on-screen in the order presented below. The session proceeded in six stages: (1) welcome and consent, (2) item preference ranking, (3) real-effort task, (4) BDM auction, (5) pain-of-paying elicitation, and (6) post-experiment questionnaire. Where screens differed by participant role or auction outcome, each variant is presented under a labeled header. Bracketed text in italics indicates procedural notes describing oTree behavior and are not participant-facing. The smiley Likert scale used in the pain-of-paying elicitation is reproduced in the main text as Figure 1; an example real-effort task matrix and the purchasable item list are reproduced within this appendix as Figure 6 and in Appendix C as Figure 8, respectively.

B.1. Welcome and Consent

Welcome

You are about to participate in an economic decision-making study. You have the opportunity to make money in this study. How much money you make depends on your decisions. Whatever money you make will be paid to you in cash at the end of the study.

During the study, verbal or nonverbal communication with other participants, cell phone use, and so on, is strictly prohibited. Communication between participants will lead to your exclusion from the study and forfeiting all monetary earnings. Please raise your hand if you have any questions; a member of the research team will come to you and answer your questions privately.

Consent

Thank you for signing up to participate in a decision-making study. The study should last about 30 minutes, and you will be paid in cash at the end of the study.

Participation is voluntary. Your decisions will remain confidential, and your name will not be recorded anywhere in the data. All information collected in this study is confidential and will be used for statistical purposes only. The data will be securely stored. The risks associated with this study are minimal. Risks are similar to everyday computer use, though some participants may feel some stress in playing a game. Although this study will not benefit you personally, the results will help our understanding of strategic behavior.

The University of Wyoming, the principal investigator, and the research team are not liable for any injury participants might sustain while participating in this study and are not able to offer financial compensation or absorb the costs of medical treatment should the participant sustain such an injury. By agreeing to participate, you do not waive any of your legal rights; however, no funds have been set aside to compensate you in the event of injury.

For additional information related to this study, contact Dr. Linda Thunström, Department of Economics, University of Wyoming, at lthunstr@uwyo.edu, or Collin DePaemelere, Department of Economics, University of Wyoming, at cdepaeme@uwyo.edu. University of Wyoming's Institutional Review Board (IRB) has determined this study to be exempt from review by the IRB administration. You can contact UW IRB if you have questions about your rights as participants (irb@uwyo.edu; 307-766-5322).

By continuing in the session, you are indicating that you are at least 18 years old, have read and understood this consent form, and agree to participate. Thank you.

B.2. Item Preference Ranking

Personal Product Ranking

Please arrange the items in the list by dragging them into **your own** preferred order. Start with the item you like the most, placing it at the top of the list (ranked as 1), and proceed to rank each item in descending order of your own preference until the item you like the least, which should be placed at the bottom (ranked as 10). All items are of equal market value and are priced at approximately \$30.

Personal Product Ranking

Please arrange the items in the list by dragging them into **your own** preferred order. Start with the item you like the most, placing it at the top of the list (ranked as 1), and proceed to rank each item in descending order of your own preference until the item you like the least, which should be placed at the bottom (ranked as 10). All items are of equal market value and are priced at approximately \$30.

1.  Coloring book with pencils	6.  Tactical flashlight
2.  Headphones	7.  Lip Balm
3.  Moisturizer	8.  Water bottle
4.  Spa kit	9.  Car detailing kit
5.  Grilling accessories	10.  Multi-tool

Figure 5: Example product ranking screen shown to participants.

Partner Product Ranking

Please arrange the items in the list by dragging them into **your partner's** preferred order, based on your understanding of their tastes. Start with the item you believe your partner likes the most, placing it at the top of the list (ranked as 1), and proceed to rank each item in descending order of preference until the item you believe they like the least, which should be placed at the bottom (ranked as 10). All items are of equal market value and are priced at approximately \$30.

You can earn money from this task. Your partner will also rank these items for themselves.

If you can correctly guess your partner's rankings, you earn \$1. Specifically, we will compare your guess of your partner's preferences to your partner's actual ranking. If you have accurately guessed the numerical ranking of at least 5 of these items, you will earn \$1.

[The item list was displayed as an interactive drag-and-drop interface.]

B.3. Real-Effort Task

Round Structure Instructions

This study will consist of 2 rounds. Within each round, your performance will directly impact your final earnings. At the end of the study, we will randomly draw one round to determine your final payoff.

Matrix Task Instructions and Comprehension Questions

In one part of the study, you or your partner will be randomly chosen to get the opportunity to earn money. The person selected will earn their money by completing a task that involves counting the number of 1s in matrices made up of 1s and 0s.

If you are chosen to do the matrix task to earn money, your task is to count the number of 1s in each matrix. Your pay from this task will be determined by how many matrices you correctly answer. If you answer a matrix incorrectly, you will move on to a new matrix. You will have 5 minutes to earn up to \$20 by completing up to 5 matrices.

You will each get an opportunity to practice the matrix task. Below is an example of a 5×5 matrix identical to what you would see during the task. In this example, there are **11 total 1s** that you would need to count.

0	1	0	1	0
1	1	0	1	0
0	0	0	0	1
0	0	1	1	0
1	1	0	1	0

Figure 6: Example matrix task screen shown to participants.

The following comprehension questions were displayed after the instructions. Participants saw the correct answer regardless of their response.

CQ1: The person assigned to earning money will earn money by counting the number of 1s in a matrix.

- True
- False

CQ1: The answer is TRUE. They will be presented with a series of matrices and will need to count the number of 1s.

CQ2: The person earning money will only be rewarded if they get every matrix correct.

- True
- False

CQ2: The answer is FALSE. They will be rewarded per matrix they complete.

CQ3: You and your partner will both get the opportunity to do the earnings task.

- True
- False

CQ3: The answer is FALSE. One of you will be picked at random to complete the earnings task, while the other waits.

Practice Round

Again, you or your partner will be given the chance to earn money by solving matrices. You will now be given a chance to practice the matrix task. In this practice round, both you and your partner will be given the opportunity to solve 3 matrices.

[Both participants solved 3 practice matrices. No earnings were associated with the practice round.]

Practice Round Results

You have finished the practice round. You correctly answered [X] of the 3 matrices. We will now move on to the real task.

[The number of correctly answered matrices was displayed dynamically for each participant.]

Role Assignment

[Participants were shown one of the following screens depending on their randomly assigned role.]

[Earner]:

You were randomly selected to earn money by completing the earnings task. You will have 5 minutes to earn up to \$20 by completing 5 matrices.

[Non-Earner]:

You were not selected to earn money by completing the earnings task. Please wait for further instructions.

[The Earner completed the matrix task. The session proceeded once the Earner finished all matrices or the 5-minute time limit elapsed.]

Matrix Task Results

[Participants were shown one of the following screens depending on their role.]

[Earner]:

You got [X] of the matrices correct. For every correct matrix, you receive \$4. You earned \$[Y].

[Non-Earner]:

Your partner has completed the earnings task. They earned \$[Y].

[Earnings were displayed dynamically. Participants who completed all 5 matrices earned \$20; earnings were reduced proportionally (\$4 per correct matrix) for those who did not finish within the time limit.]

Joint Account

The money you earned from the matrix task has been placed in a joint account you share with your partner. At the end of the study, the money remaining in this joint account will be split evenly between you and your partner. For example, if there were \$20 remaining in the joint account at the end of the study, you would each be given \$10.

There is \$[Y] in the joint account. In this part of the study, you or your partner will be randomly chosen to receive the opportunity to spend the money in the joint account.

The participant who is not given the opportunity to spend money will wait for further instructions.

The following comprehension question was displayed after the joint account instructions.

CQ4: What happens to the money in the joint account at the end of the study?

- The money goes to the participant who earned it through the matrix task
- The money is split evenly between both partners
- The participant chosen to spend money keeps all remaining funds

Correct answer: The money is split evenly between both partners. At the end of the study, all funds in the joint account are split evenly between you and your partner.

B.4. BDM Auction

[The BDM auction screens were shown only to the participant randomly assigned as the Spender. The Non-Spender was shown a waiting screen.]

Spending Task Instructions (Screen 1 of 4)

You have been selected to spend the money in the joint account. You will spend the money by participating in a special type of auction. This auction works differently from regular auctions you might be familiar with.

1. In this auction, you are not competing with other people. Instead, your bid will be compared with a random bid generated by the computer.
2. If your bid is greater than or equal to the computer's random bid, you win the auction. If your bid is lower than the computer's bid, you lose the auction.
3. If you win, you do not pay your own bid amount. Instead, you pay the computer's (lower) bid amount.

For example, suppose that you were asked to state your bid to purchase a red chair. Also, suppose we paid you a larger amount of \$1,000 to participate in this study. Your payment from participating in this study will be this amount minus anything you spend on the chair.

Let's say you decide to bid \$100 to purchase the red chair. Now we'll consider two example prices:

Scenario 1: The computer randomly draws a bid of \$82.

Since your bid (\$100) is **higher** than the computer's bid (\$82), you **WIN** the auction! You get to purchase the red chair, and you pay \$82 (the computer's bid), not your bid of \$100. So your final payment is $\$1,000 - \$82 = \$918$.

Scenario 2: The computer randomly draws a bid of \$110.

Since your bid (\$100) is **lower** than the computer's bid (\$110), you **LOSE** the auction! You do not get the red chair and do not pay anything. So your final earnings are \$1,000.

CQ5: In this auction, if you win, how much do you pay?

- You pay the amount the computer bid
- You pay the amount you bid
- You pay the average of your bid and the computer's bid

*Correct answer: You pay the amount the computer bid. If you win the auction, **you pay the amount the computer bid**.*

Spending Task Instructions (Screen 2 of 4): Tips for How to Bid

The best bidding strategy in this type of auction is to bid *exactly how much the item is worth to you*. Here's why:

Let's say you decide the red chair is worth \$200 to you. This means you'd be equally happy keeping your money or spending \$200 to buy the red chair. Is your best strategy bidding more than \$200, less than \$200, or exactly \$200? The answer is **exactly \$200**.

If you were to bid less than \$200 (say \$100), then the auction might be won by the computer (if its random bid was \$150, say). You would not be able to buy the red chair even though you were truly willing to pay more than \$150. You miss out on a good deal. Therefore, bidding less than your true value is not a good strategy.

If you bid more than \$200 (say \$300), you might win the auction but have to pay a higher price than your value of the red chair. If the computer bids \$250, you win the auction but

have to pay \$250, which is more than the \$200 the chair is worth to you. You end up with a bad deal. Therefore, bidding more than your true value is not a good strategy.

Let's say you choose to bid exactly \$200. If the computer bids less than \$200, you win and pay less than what the chair is worth to you. If the computer bids more than \$200, you lose, but that's okay because the price would have been too high anyway. You do not risk paying more for the good than you want to, and you will never pay more than the computer's bid anyway.

CQ6: If you think an item is worth \$50 to you, what amount should you bid in this auction?

- You should bid more than \$50
- You should bid exactly \$50
- You should bid less than \$50

*Correct answer: You should bid exactly \$50. In this auction you should bid **exactly how much you value the item**.*

Spending Task Instructions (Screen 3 of 4): Final Notes

- In this auction, you should always bid the exact maximum amount you would be willing to pay.
- You never pay more than what the computer bids.
- If you do not want the item at all, you should bid zero. Since the computer's bid is always positive, you will lose the auction and keep all your money.

CQ7: If you bid \$75 and the computer randomly selects \$60, what happens?

- You lose the auction and pay nothing
- You win the auction and pay \$75
- You win the auction and pay \$60

Correct answer: You win the auction and pay \$60. If your bid is higher than the computer's, you win the auction and pay the computer's bid.

Spending Task Instructions (Screen 4 of 4): Joint Account Context

Now that you know how this auction works, let's put this in the context of the study. You will be asked to bid on an item using money from the joint account with your partner. Imagine that you and your partner had \$1,000 in the joint account. Remember, at the end of the study, **anything that is not spent will be equally split** between the two of you. Imagine that you bid \$200 on the red chair, and the computer chooses \$200. What is the outcome of the auction and your final payment for the study?

- You **win** the auction, get the red chair, and you get **\$800** at the end of the study
- You **win** the auction, get the red chair, and you get **\$400** and your partner gets **\$400** at the end of the study
- You **lose** the auction and you get **\$800** at the end of the study

Correct answer: You win the auction, get the red chair, and you get \$400 and your partner gets \$400 at the end of the study. If you bid \$200 on the chair and the computer chooses \$200, you win the auction, paying \$200 and receiving the red chair. The money remaining in the joint account is $\$1,000 - \$200 = \$800$. At the end of the study, whatever is left in the joint account is split equally between you and your partner, so you receive the red chair and \$400, and your partner receives \$400.

Spending Task: Bid Screen

[The BDM auction screens were shown only to the participant randomly assigned as the Spender. The Non-Spender was shown a waiting screen.]

[Non-Spender] You were not selected to complete the spending task. Please wait for further instructions.

[Spender]

You have been selected and are receiving the opportunity to spend the money **you earned/your partner earned** from the matrix task. In other words, \$20 in the joint account. Remember, whatever balance is remaining in the joint account at the end of the study will be split equally between you and your partner.

It is important to our study that the item you purchase is for your own use and that, after the study, you do not compensate your partner for having bought the item.

The budget for you to spend is \$20.

You are being offered this item, whose market value is approximately \$30:

[The item selected by the algorithm was displayed with its description. The following text appeared below the item image.]

You will get the opportunity to buy this item by participating in an auction that works the same way as the practice auction you just learned about. If you win the auction, the item will be delivered to the lab and we will contact you so you can pick it up from us at a time that is convenient to you. The total amount you can spend on this task is the \$20, the money that **you earned/your partner earned** from the matrix task; in other words you can spend up to \$20 from the joint account. Remember, all money remaining in the joint account at the end of the study will be split evenly between you and your partner.

[During Round 2 only, the following message was also displayed:]

Remember, the results of the first and second joint account and auction **are completely separate**, and only one will be chosen randomly for your final payment.

Here's a reminder of the rules of the auction:

1. You will place a bid on the item, corresponding to how much you would be willing to pay for this item.
2. The computer will randomly draw a price for the item.
3. **If your bid is higher than the price** that the computer randomly selects, **you win the auction**, and you will pay the computer's chosen price to receive the item. **If your bid is lower than the price** that the computer randomly selects, **you lose the auction**, you do not pay the price and receive nothing.
4. The strategy for this kind of auction is to bid **exactly the maximum amount you are willing to pay** for the good.

The money you can spend on this item is the \$20 in the joint account.

How much would you be willing to spend on the item shown above in dollars? *(Enter your bid as a dollar value using at maximum 2 decimals)*

Round Transition

[Spender] You have finished Round 1 of the study, and we will now proceed to Round 2. We will repeat the earning and spending tasks. At the end of the study, we will randomly choose one of the rounds to determine your final payoff, including the result of the auction.

Non-Spender You have finished Round 1 of the study, and we will now proceed to Round 2. We will repeat the earning and spending tasks. At the end of the study, we will randomly choose one of the rounds to determine your final payoff.

[Participants then completed Round 2. Earning roles were reversed: the participant who earned in Round 1 waited in Round 2, and vice versa. The Spender remained fixed across both rounds. Each round used an independent joint account.]

B.5. Pain-of-Paying Elicitation

[This screen was shown to all participants — both Spenders and Non-Spenders — after both rounds were completed, but before the random round selection for payment.]

Feelings on Purchase

[Spender]:

You just attempted to purchase the following item using money from the joint account.

[Non-spender]:

Your partner attempted to purchase the following item using money from the joint account.

[The item selected by the algorithm was displayed with its description.]

We will ask you how you would feel purchasing the item at amounts ranging from \$0 to \$20. For instance, how would you feel if you were able to purchase the item for \$8 with money **earned by you**? How would you feel if you were able to purchase the item for \$8 with money **earned by your partner**? How would your feelings change if you were instead paying \$12 for the item? Please indicate your feelings using the smiley face scale shown in Figure 2 of the main text — ranging from very positive (Face A) to very negative (Face G).

Please indicate how you would feel about the purchase at the different prices, stating a letter (A–G) that best corresponds to your feelings.

Feelings on Purchase

Your partner attempted to purchase the following item using money from the joint account.



Headphones

We will ask you how you would feel purchasing the item at amounts ranging from \$0 to \$20. For instance, how would you feel if your partner were able to purchase the item for \$8 with money **earned by you**? How would you feel if your partner were able to purchase the item for \$8 with money **earned by your partner**? How would your feelings change if your partner were instead paying \$12 for the item? Please indicate your feelings using the smiley face scale above — ranging from very positive (Face A) to very negative (Face G).



Please indicate how you would feel about the purchase at the different prices, stating a letter (A-G) that best corresponds to your feelings.

Money YOU earned, from the joint account

Money YOUR PARTNER earned, the from joint account

Price	A	B	C	D	E	F	G
\$0 (Free)	☐	☐	☐	☐	☐	☐	☐
\$2	☐	☐	☐	☐	☐	☐	☐
\$4	☐	☐	☐	☐	☐	☐	☐
\$6	☐	☐	☐	☐	☐	☐	☐
\$8	☐	☐	☐	☐	☐	☐	☐
\$10	☐	☐	☐	☐	☐	☐	☐
\$12	☐	☐	☐	☐	☐	☐	☐
\$16	☐	☐	☐	☐	☐	☐	☐
\$20	☐	☐	☐	☐	☐	☐	☐

Next

Figure 7: Example pain of paying question screen shown to participants.

[Participants rated their anticipated pain of paying using the smiley face scale shown in Figure 1 of the main text (A = very positive, G = very negative). Pain of paying was rate at nine price points (\$0, \$2, \$4, \$6, \$8, \$10, \$12, \$16, and \$20) under two conditions (self-earned money and

partner-earned money), for a total of 18 ratings. Ratings were entered via a response grid displayed on screen.]

B.6. Post-Experiment Questionnaire

[The post-experiment questionnaire was shown to all participants after the pain-of-paying elicitation. It comprised three screens presented in the order below.]

Questionnaire Screen 1: Relationship and Spending Attitudes

In this final part of the study, we will ask you to answer a series of questions about yourself and your partner. These questions will help us gather additional information for our research. **All details are confidential and cannot be used to track your personal identity.**

Please rate how much you agree with the following statements:

(Response options: Strongly Disagree / Somewhat Disagree / Neutral / Somewhat Agree / Strongly Agree)

[Spender]

- I felt entitled to spend the money **earned by me** in the joint account
- I felt entitled to spend the money **earned by my partner** in the joint account.
- My partner would be upset with me if I spent a large portion of the money from the joint account in this study on an item only I would use.
- I would feel guilty if I spent a large portion of the money from the joint account in this study on an item only I would use.
- My relationship with my partner is very important to me.
- I feel very uncomfortable when my partner is upset with me.

[Non-Spender]

- I felt my partner was entitled to spend the money **earned by themselves** in the joint account.
- I felt my partner was entitled to spend the money **earned by me** in the joint account.

- I would be upset with my partner if they spent a large portion of the money from the joint account in this study on an item only they would use.
- I would expect my partner to feel guilty if they spent a large portion of the money from the joint account in this study on an item only they would use.
- My relationship with my partner is very important to me.
- I feel very uncomfortable when my partner is upset with me.

Do you live with your partner?

- Yes
- No

What is your current relationship status?

- In a relationship
- Married

How long have you been in a relationship with your partner? *(Please round to the nearest full year. For example, 2.3 years = "1 to 2 years.")*

- Less than 6 months
- 6 months to 1 year
- 1 to 2 years
- 3 to 5 years
- 6 to 10 years
- 11 to 20 years
- More than 20 years

How long do you believe your relationship with your partner will last?

- A few months at most
- Around a year

- Several years
- A lifetime

How many adults (age 18 or older) are in your household (including you)? (*Open-ended numeric response.*)

How many children (under age 18) are in your household? (*Open-ended numeric response.*)

Please rate how much you agree with the following statement using the smiley face scale shown in Figure 1 of the main text (A = very happy, G = very unhappy):

- How happy are you in your relationship?

Questionnaire Screen 2: Spendthrift–Tightwad Scale

[This screen administered the spendthrift–tightwad scale from Rick et al. (2008).]

Consider which of the following descriptions fits you better:

“I find it difficult to spend money”

OR

“I have difficulty controlling my spending”

Below, select a number from 1 to 11 to indicate where you fall on this dimension. 1 indicates “I find it difficult to spend money” 11 indicates “I have difficulty controlling my spending.”

(Scale endpoints: 1 = Tightwad [difficulty spending money]; 6 = About the same or neither; 11 = Spendthrift [difficulty controlling spending])

Some people have trouble limiting their spending: they often spend money — for example on clothes, meals, vacations, phone calls — when they would do better not to.

Other people have trouble spending money. Perhaps because spending money makes them anxious, they often do not spend money on things they should spend it on.

How well does the first description fit you? That is, do you have trouble limiting your spending?

(Response options: Never / Rarely / Sometimes / Often / Always)

How well does the second description fit you? That is, do you have trouble spending money?

(Response options: Never / Rarely / Sometimes / Often / Always)

The following is a scenario describing the behavior of two shoppers. After reading about each shopper, please answer the question that follows.

Mr. A is accompanying a good friend who is on a shopping spree at a local mall. When they enter a large department store, Mr. A sees that the store has a “one-day-only-sale” where everything is priced 10–60% off. He realizes he does not need anything, yet cannot resist and ends up spending almost \$100 on stuff.

Mr. B is accompanying a good friend who is on a shopping spree at a local mall. When they enter a large department store, Mr. B sees that the store has a “one-day-only-sale” where everything is priced 10–60% off. He figures he can get great deals on many items that he needs, yet the thought of spending the money keeps him from buying the stuff.

In terms of your own behavior, who are you more similar to, Mr. A or Mr. B?

(5-point scale: 1 = Mr. A; 3 = About the same or neither; 5 = Mr. B)

Questionnaire Screen 3: Demographics

What is your gender?

- Male
- Female
- Other *(please specify)*

What is your age? *(Open-ended numeric response.)*

Are you a college student?

- Yes
- No

What is the highest level of education you have completed?

- Some high school or less
- High school diploma or GED
- Some college, but no degree
- Associates or technical degree
- Bachelor's degree
- Graduate or professional degree (MA, MS, MBA, PhD, JD, MD, DDS, etc.)

What is your household's combined, pre-tax, annual income (including bonuses and commissions) last year in U.S. dollars?

- Under \$25,000
- Between \$25,001 and \$50,000
- Between \$50,001 and \$75,000
- Between \$75,001 and \$100,000
- Between \$100,001 and \$125,000
- Between \$125,001 and \$150,000
- Between \$150,001 and \$175,000
- Between \$175,001 and \$200,000
- More than \$200,001

B.7. Session Conclusion

[After the questionnaire, one round was randomly selected for payment. Participants were shown one of the following screens depending on their role and auction outcome. Dollar amounts shown below are illustrative examples; actual earnings varied by participant.]

[Non-Spender]:

The randomly selected round that determined your earnings was: **Round [X]**.

Your total earnings in the study are **\$(Y)**.

Thank you for participating!

[Spender — Won Auction]:

The randomly selected round that determined your earnings was: **Round [X]**.

In Round [X], the computer's random price was \$[P], and your bid was \$[B].

Since your bid was higher than the computer's price, you won the auction. You paid the computer's price of \$[P] for your item.

Your total earnings in the study (after removing your purchase) are \$[Y].

Please write down the following code on your receipt so we can ask you for more information about shipping: [CODE].

Thank you for participating!

[Spender — Lost Auction]:

The randomly selected round that determined your earnings was: **Round [X]**.

In Round [X], the computer's random price was \$[P], and your bid was \$[B].

Since your bid was lower than the computer's price, you lost the auction. You did not purchase an item, and no funds were removed from the joint account.

Your total earnings in the study are \$[Y].

Thank you for participating!

C. Purchasable Items in Experiment

C.1. Item Selection Process

Items were selected through an iterative process involving an initial survey, a subsequent revision for the general population, and a pilot-driven adjustment to item value. In the first stage, we targeted a college-student population and aimed for items priced at approximately \$20. We surveyed $N = 88$ University of Wyoming students on their preferences across a draft list of roughly 40 candidate items, asking each respondent to rate their desire for each item on a five-point Likert scale and reporting their gender and relationship status.

From these responses we identified the items with the largest gender preference differential (used to populate the male-oriented and female-oriented categories) and supplemented these with gender-neutral items whose average desirability across the full sample was closest to the mean desirability of the gender-differentiated items. This yielded an initial list of ten items valued at approximately \$20 each.

Because recruitment drew from the broader Laramie community rather than exclusively from the university, the item list was revised to appeal more broadly to general adult participants while preserving the same structure of male-oriented, female-oriented, and gender-neutral categories. A pilot session subsequently revealed that a \$20 item value produced insufficient separation between item value and the \$20 joint account budget, reducing the incentive to engage meaningfully with the BDM auction. We therefore identified new Amazon listings for the same item categories valued at approximately \$30, each with a customer review score of at least 4.5 stars. This yielded the final list of ten items shown in Appendix C.2. The presentation order of items in the experiment was fixed but was determined by randomization prior to data collection.

The item value exceeding the joint account balance (\$30 versus \$20) ensured that winning the auction required spending a meaningful share of the joint account, preserving real trade-offs regardless of money source.

C.2. Final Item List

Table 5 lists the ten items used in the experiment. Items are ordered as they appeared in the experiment's drag-and-drop ranking interface. All items were valued at approximately \$30. The gender orientation of each item reflects the design intent; gender-neutral items were selected to have low preference variation across male and female respondents in the initial survey.

Table 5: Items Used in the Experiment

#	Item	Gender Orientation
1	Coloring book with pencils	Female-oriented
2	Headphones	Gender-neutral
3	Moisturizer	Female-oriented
4	Spa kit	Female-oriented
5	Grilling accessories	Male-oriented
6	Tactical flashlight	Male-oriented
7	Lip balm	Female-oriented
8	Water bottle	Gender-neutral
9	Car detailing kit	Male-oriented
10	Multi-tool	Male-oriented

Figure 8 shows the product images displayed to participants during the ranking and auction stages of the experiment.



1. Coloring book
with pencils



2. Headphones



3. Moisturizer



4. Spa kit



5. Grilling
accessories



6. Tactical
flashlight



7. Lip balm



8. Water bottle



9. Car detailing
kit



10. Multi-tool
kit

Figure 8: Product images shown to participants during the item ranking and BDM auction stages. Images are the Amazon product listing photographs used in the experiment.

C.3. Incentivized Ranking Prediction Performance

Participants were paid a \$1 bonus if they correctly predicted at least 5 of their partner's item rankings exactly. Of the 200 participants who completed the experiment, 8 (4%) met this threshold. The low success rate likely reflects the strictness of the criterion: a pair of adjacent rankings being transposed (e.g., an item ranked 3rd predicted as 4th) counted as two misses rather than near-misses. Actual preference alignment between partners was likely higher than this figure implies. The low bonus rate does not affect the validity of the item assignment procedure: the algorithm used participants' submitted rankings regardless of prediction accuracy, and the incentive was sufficient to motivate genuine effort even if few participants met the exact threshold.

D. Additional Results

Supplementary figures for Section 5 are collected here. Figure 9 and Table 6 provide the full sample robustness checks for the pain of paying result; Figure 10 provides the between-subjects robustness check for the willingness to pay result; Figures 11 and 12 present the subset and correlation analyses from Sections 5.1 and 5.2; Figures 13 to 15 present heterogeneity subgroup distributions from Section 5.3; and Figure 16 presents power curves from Section 5.4.

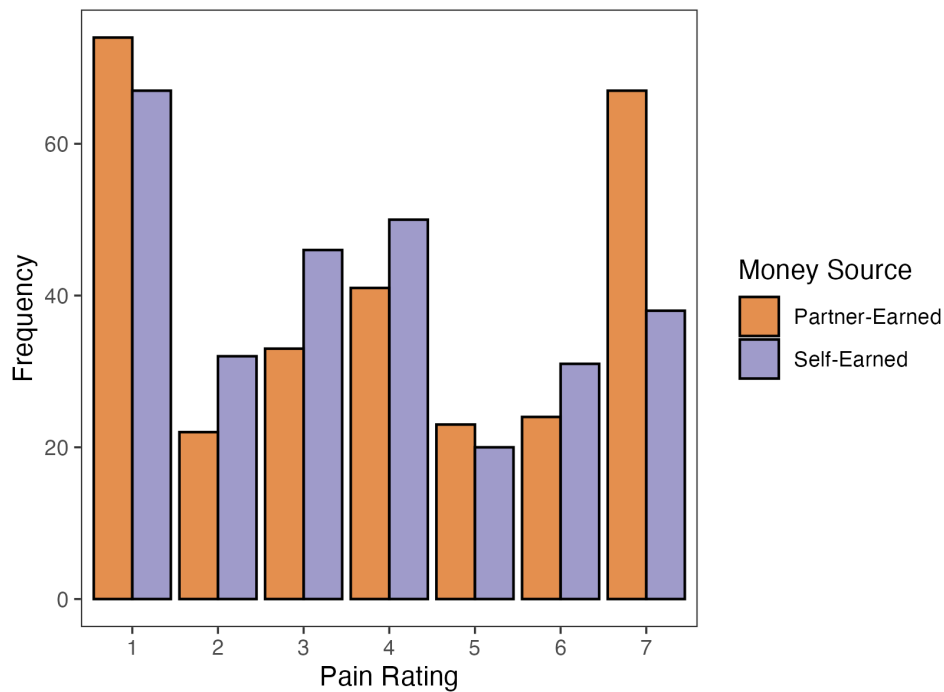


Figure 9: Pain-of-paying distributions at the \$10 price point for the self-earned and partner-earned conditions ($N = 200$ participants). The scale runs from 1 (no pain) to 7 (maximum pain).

Table 6: Pain of Paying Ratings For Full Sample Across Price Points

Price (\$)	Mean Self	SD Self	Mean Partner	SD Partner	Mean Diff
0	1.09	0.60	1.10	0.65	-0.01
2	1.53	1.26	1.81	1.51	-0.28
4	1.88	1.39	2.25	1.76	-0.38
6	2.37	1.65	2.74	1.99	-0.37
8	2.88	1.81	3.24	2.10	-0.36
10	3.55	2.03	3.90	2.22	-0.35
12	4.33	2.00	4.47	2.25	-0.14
16	5.00	1.95	4.98	2.21	0.02
20	5.43	1.93	5.27	2.18	0.16

Notes: $N = 200$ participants from 100 couples. Mean pain ratings on the smiley Likert scale (1 = no pain, 7 = maximum pain) at each price point under the self-earned and partner-earned conditions. Mean Diff = Mean Self – Mean Partner. The \$10 price point is the primary outcome; the \$2 price point is used in the robustness check in the text. At price points \$16 and \$20, the partner-earned mean falls slightly below the self-earned mean.

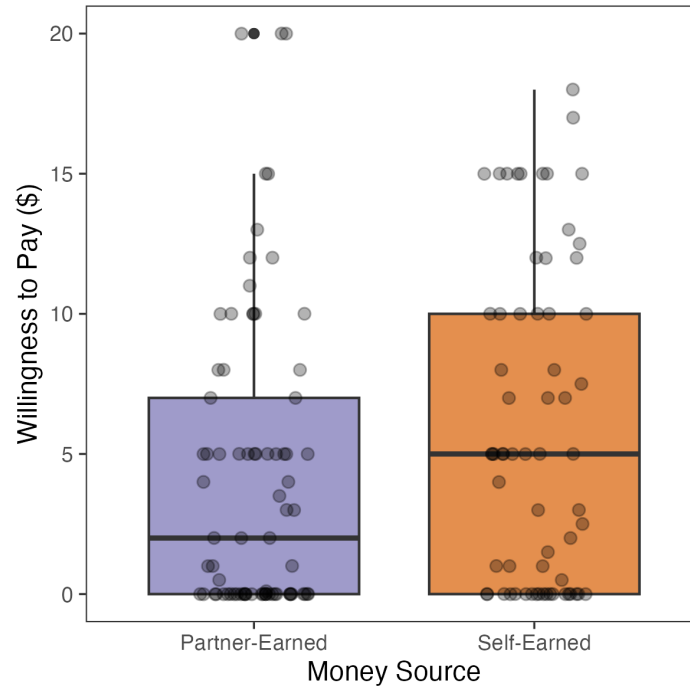


Figure 10: Willingness to pay distributions by money source condition among Round 1 spenders only, providing a between-subjects comparison unaffected by within-subject contamination. Self-earned: $N = 44, \mu = \$5.67 (SD = \$5.64)$. Partner-earned: $N = 56, \mu = \$4.02 (SD = \$5.58)$.

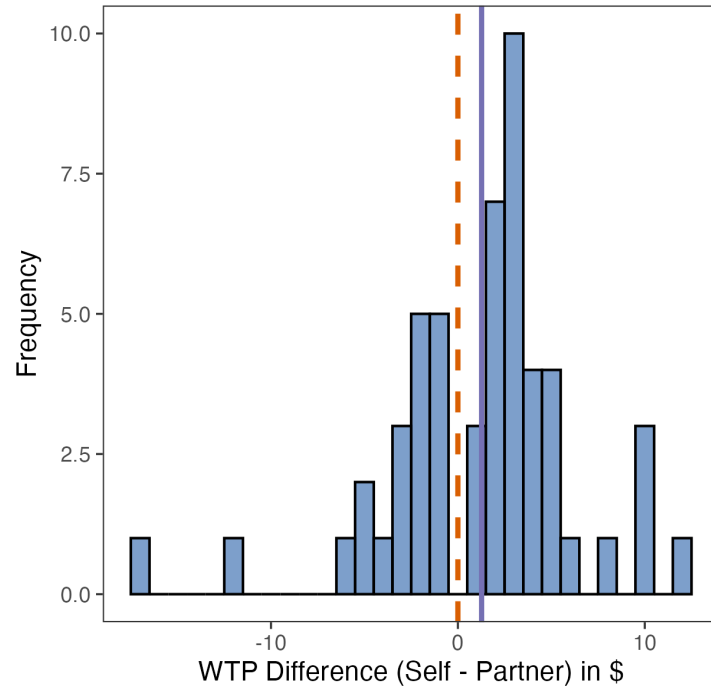


Figure 11: Distribution of within-subject willingness to pay differences among the 39 spenders who stated different bids across conditions (the Different and One Zero groups). The vertical blue line marks the sample mean (\$1.24); the dashed orange line marks zero.

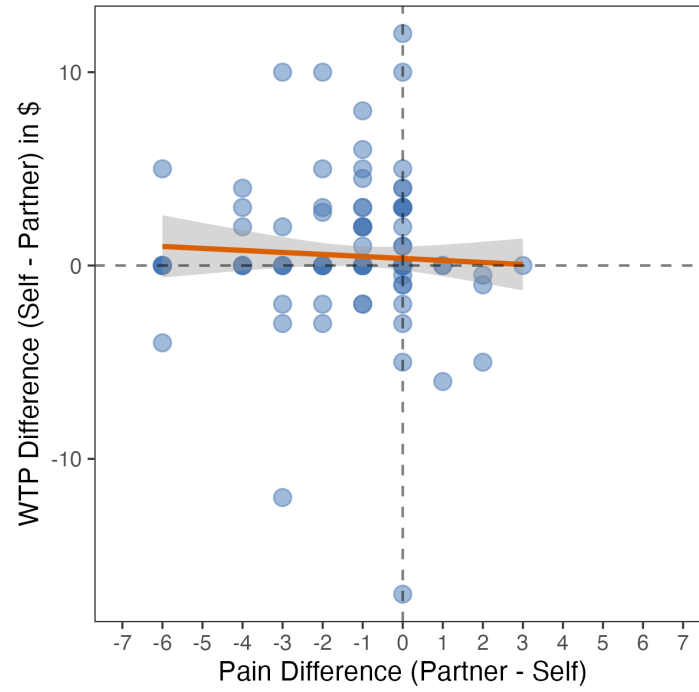


Figure 12: Scatterplot of individual willingness to pay differences (Self–Partner) against pain-of-paying differences (Partner–Self) at the \$10 price point ($N = 100$ spenders). The orange line is a locally smoothed fit. Spearman’s $\rho = 0.144$, $p = 0.154$.

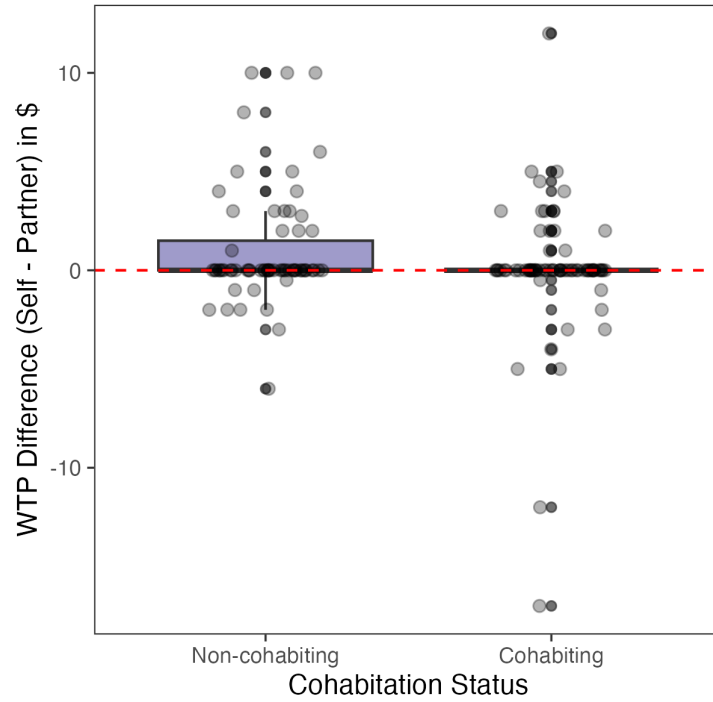


Figure 13: Within-subject willingness to pay differences by cohabitation status. Non-cohabiting: $N = 45, \mu = \$0.78$. Cohabiting: $N = 55, \mu = \$0.24$. Red dashed line marks zero. Results are exploratory.

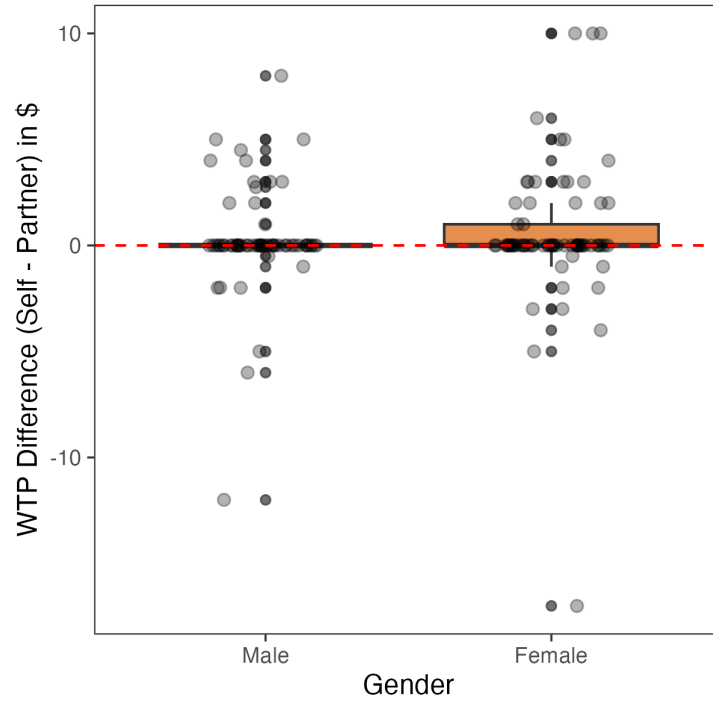


Figure 14: Within-subject willingness to pay differences by spender gender. Male: $N = 47, \mu = \$0.19$. Female: $N = 49, \mu = \$0.62$. Non-binary spenders ($N = 4$) are excluded from this figure. Red dashed line marks zero. Results are exploratory.

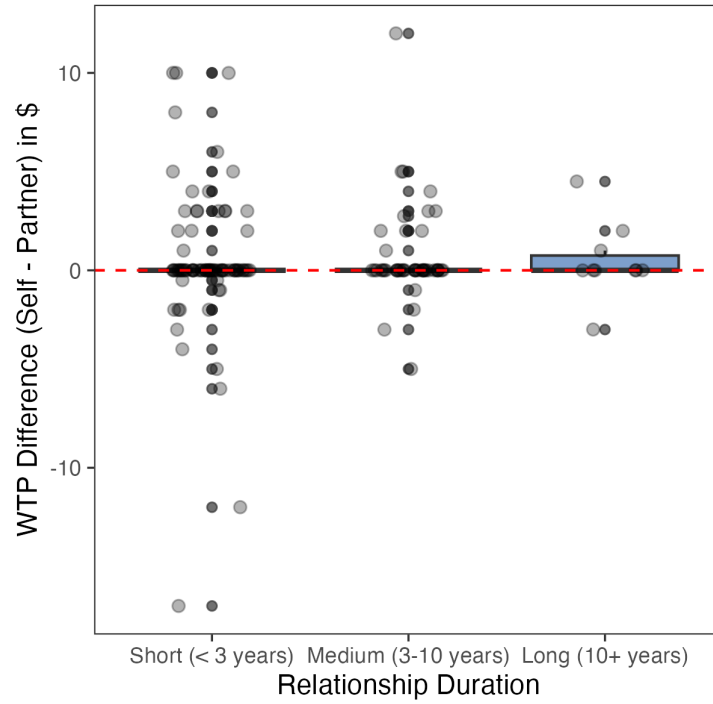


Figure 15: Within-subject willingness to pay differences by relationship duration category. Short (< 3 years): $N = 56$, $\mu = \$0.41$. Medium (3–10 years): $N = 37$, $\mu = \$0.59$. Long (> 10 years): $N = 7$, $\mu = \$0.50$. Red dashed line marks zero. Results are exploratory.

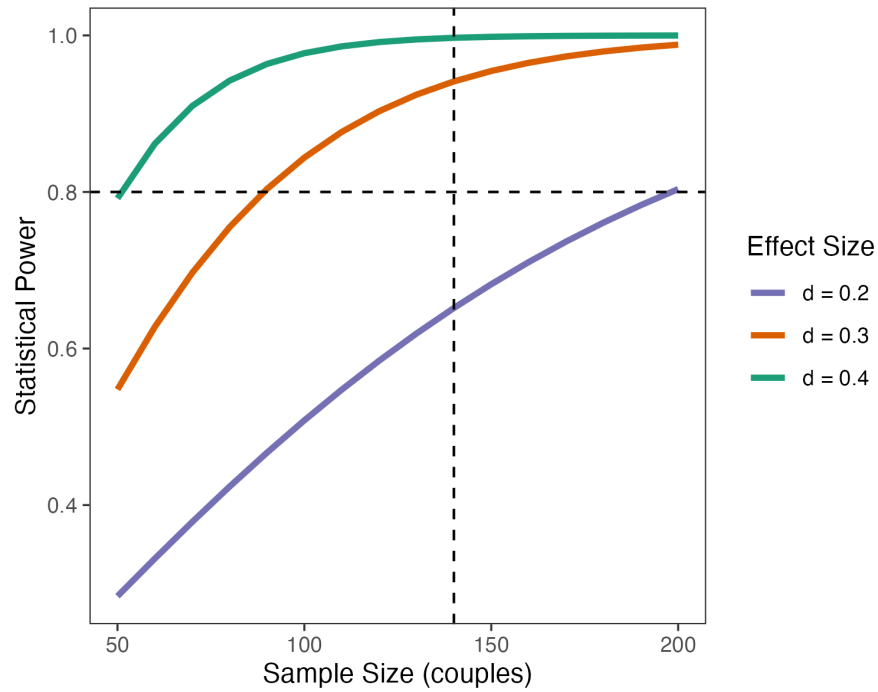


Figure 16: Post-hoc power curves for a paired t -test at effect sizes $d = 0.2, 0.3$, and 0.4 . The vertical dashed line marks the target sample of 140 couples; the horizontal dashed line marks 80% power. The observed willingness to pay effect $d = 0.087$ falls below the plotted range.